Type MS · Compare light and electron microscopes and use their parts

Answer 1 of 6 · Question 3 part c · 2 marks

2023 JUN · Paper 02 (regional R) · find part c in this mark scheme extract

3 (a)	• contains particular features / characteristics/ adaptations; • to perform a specific function;		2
(b) (i)	• 84 ÷ 0.006; • 14000x	Allow correct final answer from a calculation that includes 84 and 0.006, max 1 mark	2
(ii)	half the amount/haploid/23 chromosomes;		1
(iii)	• to provide energy; • for swimming/movement/travelling;		2
(iv)	• 180 ÷ 0.070; • 2571; • (÷ 60) 42.85;	Allow any figure that rounds to 43 Allow ecf for third mark - an incorrect value for a calculation carried out for mps1 and 2, divided by 60 **and** a correct answer to that calculation	3 exp
(c)	Any two from: • greater magnification; • higher resolution/can see objects more clearly/in more detail; • can see a greater number of structures;		2 exp

Total Question 3 = 12 marks

Type MS · Compare light and electron microscopes and use their parts

Answer 2 of 6 · Question 2 part a.ii · 1 mark

2019 JUN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

2 (a)	(i)	<table><tr><th>Part</th><th>Name of part</th><th>Function</th></tr><tr><td>X</td><td>mitochondria;</td><td></td></tr><tr><td>Y</td><td></td><td>controls activities of the cell/stores DNA / genetic information;</td></tr></table>		1 1
	(ii)	**A,** more cell structures can be seen with greater resolution		1
	(iii)	60mm/6cm; 60 ÷ 0.05; 1200;	Ecf two marks for correct calculation from incorrect measurement Full marks for correct final answer	3
(b)		• diffusion; • from a high concentration (in the cell) to a lower concentration (in the blood)/down a concentration gradient;		1 1

Total question 2 = 8 marks

Type MS · Compare light and electron microscopes and use their parts

Answer 2 of 6 · Question 2 part a.ii · 1 mark

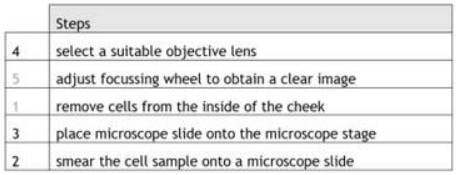
2019 JUN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Compare light and electron microscopes and use their parts

Answer 3 of 6 · Question 3 part a.ii · 1 mark

2020 NOV · Paper 01 (regional R) · find part a.ii in this mark scheme extract

Number				
3	(a) (i)		One mark for 4 before 5. One mark for 3 before 2	2
	(ii)	more detail/organelles can be seen/organelles/named organelle seen more clearly (1)		1
	(iii)	reference no stain applied to specimen(1)		1
	(iv)	• reference to sterilising (named) equipment (1) • wash hands to avoid cross contamination(1) • incubate/store in sealed petri dish / container	allow valid sterilisation technique	2
	(b) (i)	10 × 40 = 400	Full marks for correct answer without working	1
	(ii)	reference to 1000 (1) 0.018 × 1000 = 18 (µm)	Full marks for correct final answer	2

Total Question 3 = 9 marks

Type MS · Compare light and electron microscopes and use their parts

Answer 4 of 6 · Question 5 part a.ii · 2 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

number				
5	(a) (i)	• join with ovum; • during fertilisation;		2
	(ii)	any two from • ribosomes; • endoplasmic reticulum/ER; • chromosomes;		2
	(b)	• site of (aerobic) respiration; • energy released/ATP produced; • sperm motile/active/swim;		3
	(c)	3µm = 0.003 mm; distance between X and Y = 145 mm; 145; 0.003 =48333/48000;	ECF = 2	4

Total 11 marks

Type MS · Compare light and electron microscopes and use their parts

Answer 4 of 6 · Question 5 part a.ii · 2 marks

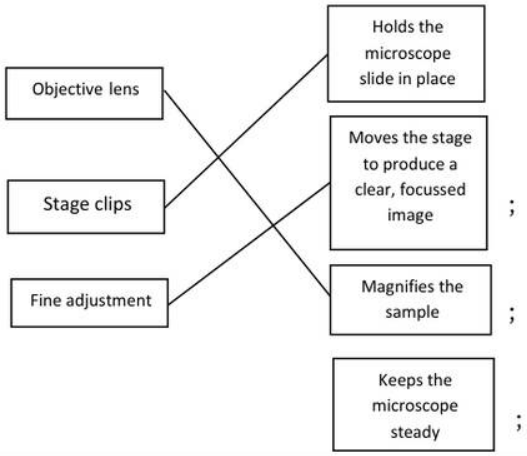
2021 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Compare light and electron microscopes and use their parts

Answer 5 of 6 · Question 1 part b.i · 3 marks

2024 JUN · Paper 01 (regional R) · find part b.i in this mark scheme extract

Number			
1 (a) (i)	**Any two from:** • (both) have a cell membrane; • (both) have cytoplasm; • (both) have genetic material/DNA; • (both) contain ribosomes;		(2)
(a) (ii)	**Any two from:** • bacterial cell has a cell wall / an animal doesn't; • bacterial cell has no nucleus /an animal cell does; • bacterial cell has a plasmid /the animal cell doesn't; • bacterial cell has a flagellum/ the animal cell doesn't; • bacterial cell has pili /the animal cell doesn't; • bacterial cell has no mitochondria /the animal cell does; • bacterial cell has slime capsule / the animal cell doesn't;		(2)
(b) (i)	 ; ; ;	Reject more than one line from each part	(3)
(b) (ii)	• substitution, 5 ÷ 50 000 = 0.0001(cm); • conversion to mm - 0.0001 x 10; • 0.001 mm/1×10^{-3}; or • 5×10=50mm; • 50÷50000; • 0.001mm/1×10^{-3}	ecf for mp 2 and 3 Full marks for correct final answer	(3)

Total for Question 1 = 10 marks

Type MS · Compare light and electron microscopes and use their parts

Answer 6 of 6 · Question 7 part c · 2 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

7 (a)

Part	Name of part	Function of part
X	cell membrane;	controls what enters/leaves cell;
Y	nucleus;	controls cell/stores genetic information;
Z	cytoplasm;	site of chemical reactions;

If they get any wrong in the part column, they can score marks for the correct function.

6

(b)

- scrape inside of cheek/cotton bud to remove cheek cells/eq.;
- place (scraping) on slide;
- smear/spread thinly;
- add stain/ methylene blue;
- place coverslip on top;
- remove excess stain;
- sterile/disinfect cotton bud;

4 (

(c)

- ribosomes;
- mitochondria;

accept any suitable structures such as ER, golgi, RER

ignore chromosomes

2

Total = 12

PAPER TOTAL - 90 MARKS

Type MS · Compare light and electron microscopes and use their parts

Answer 6 of 6 · Question 7 part c · 2 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

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